

Reproducible research through reusable code

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What is reproducibility?

		Data	
		Same	Different
Analysis	Same	Reproducible	Replicable
	Different	Robust	Generalisable

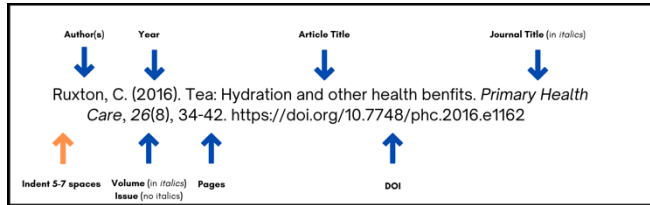
Source: [the Turing Way](#)

- Our focus today is on **computational reproducibility**:
 - Is there enough detail to repeat the same study?
 - Availability of data, code, and methods make recomputation possible

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A small thought experiment

- Think about the role of bibliographic citations in research
 - Bibliographic citations are long established: formal structured references to another scholarly work



Example APA citation from [Victoria University Library Guides](#)

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A small thought experiment

- Imagine we replace literature references with casual mentions, in varying, unpredictable, and incomplete formats
- "...some article from some well-known professor showed X"
- "a paper by Y and colleagues in a high-impact journal..."
- "Johnson (2024), *Nature*"



Photo by [Hans-Peter Gauster](#) on [Unsplash](#)

Why is reproducibility important?

- Claim of the day: Showing as detailed as possible how certain results were calculated should be as basic as correctly referencing literature

“An article about computational science [...] is merely advertising of the scholarship. The actual scholarship is the complete software development environment and the complete set of instructions which generated the figures.”

Buckheit & Donoho (1995) after Claerbout



Photo by [Fox](#) on [Unsplash](#)

But to do this well, is arguably quite hard

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Software code is essential for science

- Computational methods ubiquitous and have transformed science
- But methods mostly discussed informally in papers
- Code that produced the results is usually not available
- Data & software are:
- An essential part of research
- First-class research objects



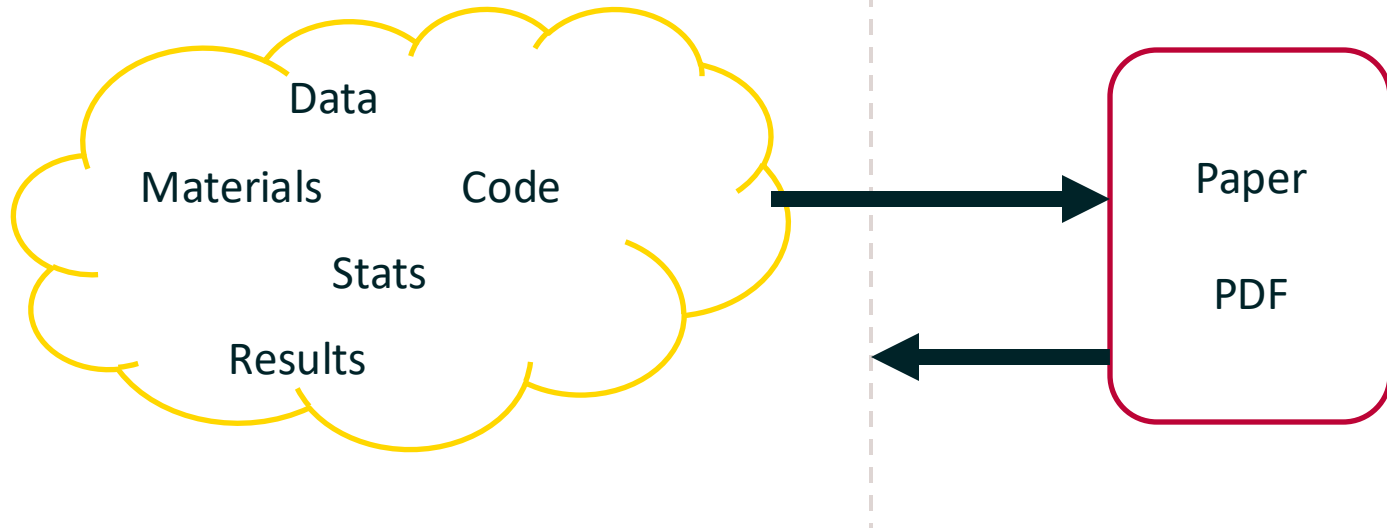
Photo by [Sean D](#) on [Unsplash](#)

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Sharing the actual work versus the pdf

- IN THE LAB

OUTSIDE



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Why is reproducibility important?

- Independent recomputation
→ increased trust in results (minimum condition for finding to be believable / informative)
- Prerequisite for reproducibility is full transparency of code
→ allows for better understanding, interpretation, and re-use
- Reproducible workflows will make your life easier
→ easier to correct, update, continue your work



Photo by [Proxyclick Visitor Management System](#) on [Unsplash](#)

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Working reproducibly is not trivial

- Takes time to learn new tools and to document your process as you go
- Documentation is key: number of lines of documentation may exceed your code
- But *partially reproducible* is better than *not at all reproducible*. Just try to make your next paper or project better organized than the last.



Photo by [Hayley Murray](#) on [Unsplash](#)

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Improving computational practices and code sharing

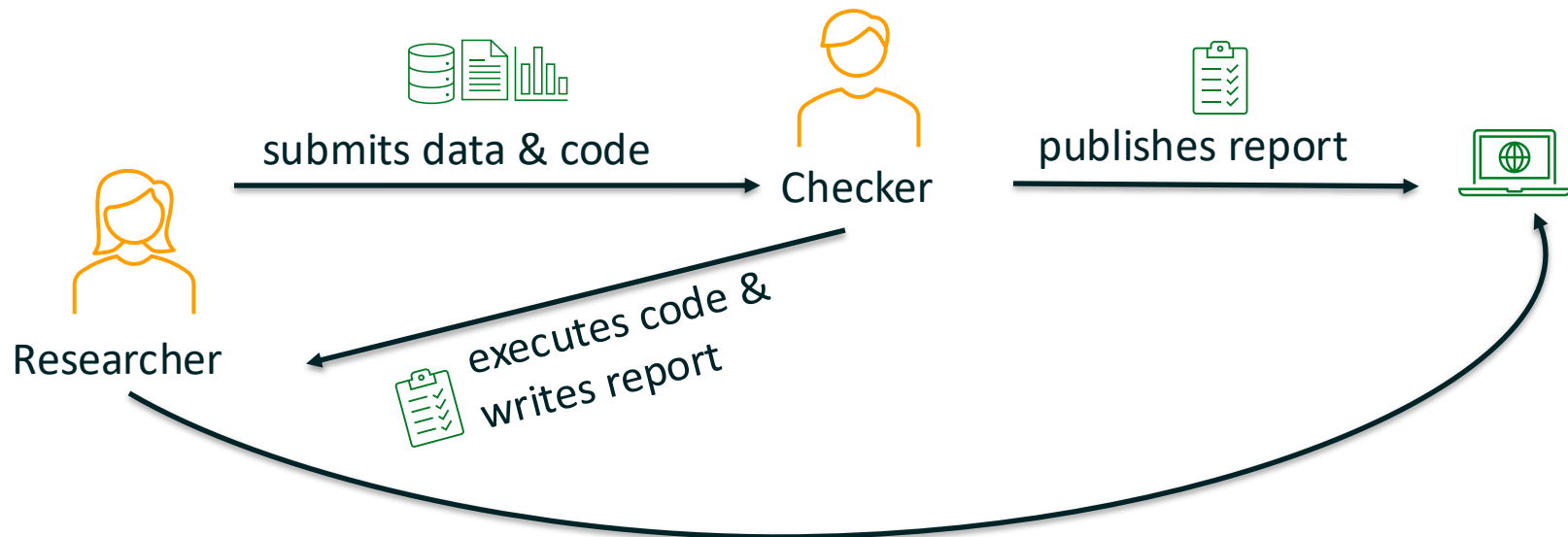
- Best practices for scientific computing are not widely adopted



- Focus on basic, computational reproducibility (same data and code reproduce same results)

Reproducibility checks to improve computational practices

- Aim: develop a workflow to provide a reproducibility check for data and code submitted alongside research publications



Similar initiatives:

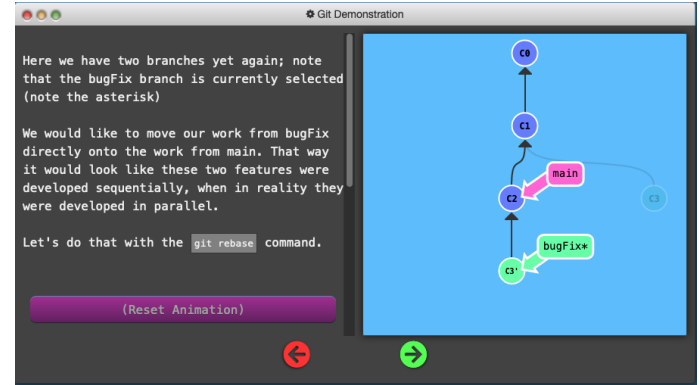


can reference report
in publication

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Use tools for version control

- Data and analysis code usually change over time
- How to reproduce a particular figure or result?
- Facilitated when data, code and figure files are under versioning
- Comments in version history help us understand what changes were made, or why a specific analysis was run

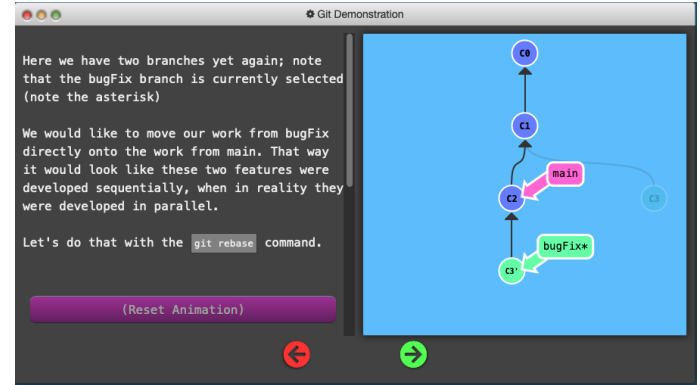


Picture: from [Learn Git Branching interactively](#)

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Use tools for version control

- Versioning: creating multiple versions (manually)
- Version control: systematic approach to versioning
 - Preferable when a lot of files are changing, like when one creates code or work with data with multiple collaborators

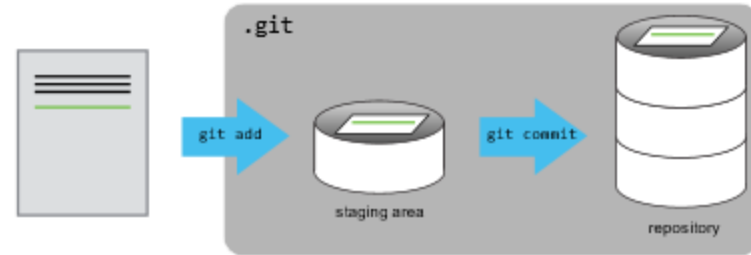


Picture: from [Learn Git Branching interactively](#)

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Use tools for version control

- Git is free and open source
 - Allows multiple local branches for parallel development
- Files are kept in a repository
 - users clone copies of the repository for editing
 - commit changes back to the repository when they are done

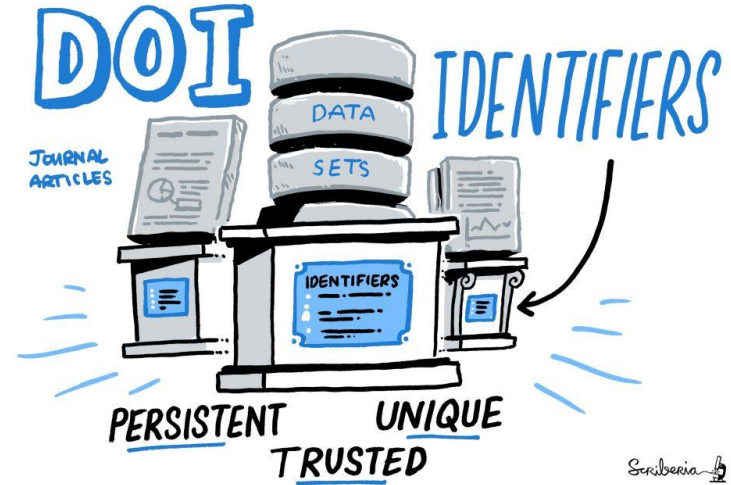


Picture: from [Software Carpentry: Version control with Git](#)

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Publish your code

- Publish your code in a repository
 - Your project will receive a persistent identifier
 - You can easily publish a [GitHub repository on Zenodo](#)



The Turing Way project illustration by Scriberia. [Zenodo](#)

Ezra

Program

09:30	Welcome & icebreaker
09:45	Workshop introduction: why is reproducibility important?
10:15	Software dependencies
10:30	Break
10:40	Software dependencies continuation
11:30	Break
11:40	Code conventions and modular coding
12:30	Lunch
13:30	Next steps: how to make your code reusable?
13:45	Work on your own project
15:00	Break
15:10	Reusability check
16:15	Wrap-up
16:30	DRINKS

A stylized, handwritten-style logo for Erasmus, featuring a large, flowing 'E' followed by the word 'Erasmus' in a cursive script.